

Chapter 4 Activity Based Costing

Activity 1: Job cost with Departmental Overhead rates

Neta Corporation has the following 2 departments:

	Assembly Department	Machining and Painting Department	Total
Total Budgeted Overhead (MOH)	\$500,000	\$2,000,000	\$2,500,000
Budgeted DL Hours (DLH)	200,000	50,000	250,000
Budgeted Machine Hours (MH)	20,000	120,000	140,000

The assembly department is more labor intensive, while the Machining and Painting Department has more equipment and thus higher maintenance, power consumption, depreciation, and set-up costs. Therefore, using department rates, the Assembly Department would use DLH to allocate its overhead while the Machining and Painting department would use MH.

1. Compute the pre-determined MOH Rate for the PLANT assuming that the company assigns all MOH based on DLH.

$$\text{\$2,500,000 Total MOH} / \text{250,000 Total DLHs} = \text{\$10/DLH}$$

2. Compute the pre-determined overhead rates for the Assembly department and the Machining and Painting Department.

Assembly Dept.:

Mach. & Paint Dept.:

The company has Job #13 with the following information:

	Assembly Department	Machining and Painting Department	Total
Total Job #13 (14,400 units)			
Prime Costs (DM + DL)	\$100,000	\$20,000	\$120,000
Direct Labor Hours	5,000	1,000	6,000
Machine Hours	200	500	700

3. Using plantwide overhead rates, develop a per-unit cost for job #13.

Prime Costs

+MOH Applied

=Total Job Costs

Divide by # of Units

Per Unit Cost

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4. Using departmental overhead rates, develop a per-unit cost for the job #13.

Prime Costs

+ MOH Applied Assembly

+MOH Applied Mach & Paint

=Total Job Costs

Divide by # of Units

Per Unit Cost

5. Which cost (plantwide or departmental) do you think is more accurate? Why?

6. Explain whether or not a cost measurement system can impact how profitable a company is?

Note:

1. Does the nature of production affect whether we should use Activity Based Costing or Normal costing?
When a company produces multiple products, and they don't consume resources uniformly, then applying a uniform rate for MOH becomes redundant. Activity Based Costing provides a better match between costs incurred and the activities that drive them. If products are uniform, then the single plantwide rate yet makes the best sense to use.
2. How to apply ABC?
Same process as Normal Costing. Instead of finding a single uniform plant wide rate, we will now treat each activity as a separate cost pool and find a cost driver for each pool. Note that we can have different cost drivers for each activity pool based on what drives the incidence of that cost. For each activity or cost pool, we determine the estimated overhead rate. Next we apply that rate to the actual activity or cost driver for that pool to arrive at Applied MOH. We will do this for as many pools we have identified and add each of the Applied MOH arrived at for each pool. The total we arrive at, is the total Applied MOH for the product.

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Activity #2 = Classifying costs within the cost hierarchy

Classify each of the following costs as either unit-level, batch-level, product-level or facility-level.

	Unit Level	Batch Level	Product Level	Facility Level
1. Product line manager salary				
2. Machine setup costs that are incurred whenever a new production order is started				
3. Patent for new product				
4. Factory utilities				
5. Direct Materials				
6. Cost to inspect each unit of product as it is finished				
8. Depreciation on factory				
9. Direct Labor				
10. Costs to purchase raw materials of products.				

Note:

Unit level cost – occurs every time one unit of the product is manufactured, typically Direct Material, Direct Labor.

Batch level cost – occurs every time a batch is processed, typically, batch set up costs, or cleaning costs between two batches.

Product level cost – occurs every time a “product line” is impacted. Make sure you distinguish between “unit” and product. You could have thousands of units within one product line. Examples include product design cost.

Facility level costs – Factory costs that do not belong to any of the above categories.

Some costs could be classified into more than one category.

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Activity #3 - Activity rates and cost of a unit using Activity-Based Costing

Courtney Corporation manufactures two products—Tricycles and Wagons. The annual production and sales of Tricycles is 2,000 units, while 1,500 units of Wagons are produced and sold. The company has traditionally used direct labor hours to allocate its overhead to products. Here is the data related to the company's two products:

	Tricycles	Wagons	
Direct Materials per unit	\$ 20	\$ 15	
Direct Labor \$ per unit	\$ 10	\$ 10	
DLH per unit	1.5	1	
Annual Production	2,000	1,500	
Total DLHs	3,000	1,500	4,500

The company is looking at the possibility of changing to an activity-based costing system for its products. If the company used an activity-based costing system, it would have the following three activity cost pools:

Activity cost pool	Cost Driver	Estimated overhead cost	Expected activity		
			Tricycles	Wagons	Total
Setup costs	# of setups	\$50,000	25	75	100
Engineering costs	# of engineer hours	\$7,500	60	40	100
Maintenance costs	# of machine hours	\$157,500	3,000	1,500	4,500
Total MOH		\$215,000			

Plantwide:

Predetermined overhead rate under the traditional system based on a single plantwide overhead rate =

PDOR: Total Est MOH/Total Est. DLH =

Cost of one Tricycle and one Wagon under **traditional (plantwide)** system

Tricycles:

Wagons:

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1. Calculate the unit cost for a Tricycle and Wagon using the activity-based costing system.

Activity cost pool	(a) Total cost	(b) Total activity	(a)/(b) Activity rate
Setup costs	\$50,000	100	
Engineering costs	\$7,500	100	
Maintenance costs	\$157,500	4,500	

Activity cost pool	(c) Activity rate	(d) Total activity	(c)x(d) Activity-based cost
Tricycles			
Setup costs			
Engineering costs			
Maintenance costs			
Total MOH Applied			
Divide by # Units			
= Overhead cost per unit			
Direct Materials per unit			
Direct Labor per unit			
Total Cost Per Unit			
Wagons			
Setup costs			
Engineering costs			
Maintenance costs			
Total MOH Applied			
Divide by # Units			
= Overhead cost per unit			
Direct Materials per unit			
Direct Labor per unit			
Total Cost Per Unit			

2. What happened to the cost of Tricycles and Wagons under ABC compared to plantwide? Why?

Note:

When a company manufactures multiple products which do not consume resources evenly, using a single plant wide rate will result in a cost distortion of allocated overheads. One product will be undercosted while the other will be overcosted. Total OH allocated will be the same, but the proportion in which they are allocated to each product is likely to be inaccurate.

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Activity #4 - Compare ABC to Plantwide cost allocation

Kate Systems produces two different products. Below is the data related to the company's two products:

	Product A	Product B
Sales Price per unit	\$ 1,000	\$ 2,000
Direct Materials per unit	\$ 500	\$ 750
Direct Labor per unit	\$ 200	\$ 350
Number of Units	10,000	7,500

Part A (Traditional Costing-Plantwide rate)

Kate currently uses a normal job order costing system with estimated total manufacturing overhead of \$6,800,000 applied to the two products on the basis of machine hours. Total machine hours estimated to be 8,000.

1. Compute the current overhead rate.
2. Determine the indirect costs (MOH) applied to Product A if Product A uses 2,000 machine hours. Calculate the total MOH applied and MOH applied per unit for Product A.
3. Determine the indirect costs (MOH) applied to Product B if Product B uses 6,000 machine hours. Calculate the total MOH applied and MOH applied per unit for Product B.
4. Determine the Gross Profit Dollars & Gross Profit Percentage per unit and in total for Product A and Product B

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Part B (Activity Based Costing): Kate is considering implementing an ABC system and determined the determined the following are the three primary activities: Design, which uses Engineering Hours as an activity driver; Machining, which uses machine hours as an activity driver; and Inspection, which uses number of batches as an activity driver. The cost of each activity and usage of the activity drivers are as follows:

	Cost	Usage by Product A	Usage by Product B
Design (Engineering Hours)	\$1,500,000	400	600
Machining (Machine Hours)	\$5,000,000	2,000	6,000
Inspection (Batches)	\$300,000	90	30

Required:

6. Calculate the activity rate for Design.
7. Calculate the activity rate for Machining.
8. Calculate the activity rate for Inspection.
9. Determine the indirect costs (MOH) assigned to Product A in total and per unit

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10. Determine the indirect costs (MOH) assigned to Product B in total and per unit

11. Determine the Gross Profit Dollars & Gross Profit Percentage per unit and in total for Product A

12. Determine the Gross Profit Dollars & Gross Profit Percentage per unit and in total for Product B.

13. The President of Kate is skeptical about whether the benefits of ABC will be worth the cost. She notes that ABC is just a different way of “spreading the same total of cost around”, so overall company profitability will remain the same under either method. Do you agree with this view? Why or why not?

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Activity #5 - Identifying Costs as value-added or non-value added

Indicate whether each activity is value added or non-value added in a manufacturing process.

Product design research	_____
Moving materials	_____
Machining	_____
Assembly of components	_____
Finished goods inventory storage	_____
Rework after a quality inspection	_____
Painting product	_____

Activity #6 - Classifying Costs of Quality

Indicate how each of the following costs would be classified as in regards to quality:

Prevention, Appraisal, Internal Failure, External Failure

Costs incurred to inspect products after manufacturing. _____

Costs incurred to handle customer complaints. _____

Costs incurred to train production employees. _____

Costs incurred for downtime of production equipment. _____

Note:

Value Added activities that add “value” to the finished unit of production. Not all “non- value added” activities can be eliminated.

Costs of quality are those costs that will be incurred by every company that produces units. Management would like to position itself to incur costs that are typically not customer facing – because this leads to loss of reputation and market share. They would rather incur higher prevention, appraisal and internal failure costs in that order to minimize external failure costs.